# Kenneth Cao

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## About Me

Final-year Computer Science student at the University of Bath with hands-on experience building full-stack web applications during a 13-month placement at Mayden. Passionate about solving real-world problems through clean code and thoughtful design. Proven ability to work effectively in agile teams and independently as shown in multiple academic projects with strong results. Seeking opportunities to contribute to impactful software engineering projects.

## Education

**University of Bath**, BSc (Hons) in Computer Science with Professional Placement — Sept 2023 – July 2027

- **Second Year Average: 68.33%**
  - Modules: Advanced Programming, Algorithms and Complexity, ML, Cybersecurity, Visual Computing, HCI
- **First Year Average: 75.17%**
- Modules: Programming I & II, AI I, Discrete Maths and Databases, Systems Architecture, Maths for Computation

**Aquinas College** — Sept 2021 – July 2023

- A* in A Level Maths
- A* in A Level Further Maths
- A* in A Level Computer Science

**St Anne's RC Voluntary Academy** — Sept 2016 – July 2021

Achieved seven grade 9s in and three grade 8s in GCSEs.

## Skills

- **Languages:** PHP, JavaScript, TypeScript, Python, C#, Java, Rust, Haskell
- **Frameworks:** Laravel, Slim, React, PyTorch
- **Databases:** MySQL, SQLite
- **Tools:** Git, Docker, Vim, VS Code, Jetbrains IDEs

## Experience

**Placment Fullstack Software Engineer**, Mayden — July 2025 – August 2026

Worked in a self-managed Agile Scrum environment where I developed new features and value for Mayden's EPR solution where accessibility and security were paramount. Utilised technologies such as PHP, Javascript, React and MySQL where I would not only work fullstack to build the backend architecture as well as customer facing frontend features.

**Game Developer**, BlackBox Studios — July 2024 – September 2024

Developed physics systems and collision detection mechanics for BoomBall Extreme in Unity and C#, ensuring responsive gameplay across various object interactions. Showcased at EGX London 2024 and Protoplay.

## Projects

**Rust Raytracer**

Engineered a custom ray tracer in Rust by translating and adapting foundational C++ rendering algorithms from the "Ray Tracing in One Weekend" series. Leveraged Rust's ownership model and memory safety principles to develop a highly performant rendering engine.

**Birdography**

Collaborated on a year-long Software Engineering project to develop an Android app in Kotlin that motivates users to engage with nature by birdwatching. Implemented ML-based image recognition to identify birds from user photos and enable collection-building features.

**Email Spam Filter**
Engineered a custom neural network from scratch using Python and NumPy to classify emails as spam or legitimate, implementing training and classification algorithms independently without high-level frameworks like PyTorch or Keras.

**Best Use of AI Winner at Bath Hack 2024**
Developed a Chrome extension for a hackathon that uses AI to let users virtually "try on" clothes found online by generating a 3D model based on their body measurements or uploaded images. My role focused on integrating the Stable-Diffusion-webgui API to apply clothing to the 3D models.

**First Place Winner at Physics Society Hackathon 2024**
Analyzed black hole data to investigate differences between BALs and regular quasars. Processed datasets using Haskell and generated Python visualizations to identify correlations between redshift, mass, and luminescence of regular quasars and BALs.

**Personal Informatics system to manage student budgets**
Collaborated in an agile team to develop a student-focused Personal Informatics system for a Software Engineering project. Led front-end GUI development and data visualization, and rotated as project manager and scrum master throughout the project.

**Maze Crawler Game**
I made this game using the monogame framework in C# and utilises a recursive backtracking algorithm in order to generate a unique maze for the map. I also utilised an A* algorithm in order for the monsters that spawn in the game to track and follow the player after they are within proximity of each other.

## Volunteer Roles

**General Committee Member**, Bath Computer Science Society — April 2025 – July 2027
Directed the planning and execution of large-scale society events, most notably a flagship hackathon yielding 220+ attendees and multiple industry sponsorships. Cultivated a vibrant departmental community by orchestrating supplementary events such as game jams and social gatherings.

**Chef**, Vegetarian Society — April 2026 – July 2027
Orchestrated weekly dining events for the Vegetarian Society catering to over 40 attendees. Managed end-to-end logistics, including strict budget oversight, menu planning, and meticulous adherence to 5-star food hygiene protocols.

**Peer Assisted Learning Leader**, University of Bath — September 2024 – July 2025
Coordinated and facilitated academic support sessions for first-year Computer Science students. Collaborated with co-leaders to deliver structured study materials, honing strong organizational, public speaking, and presentation skills.

**Peer Mentor**, University of Bath — September 2024 – July 2025
Facilitated the onboarding and integration of first-year students by providing comprehensive mentorship. Delivered practical guidance on navigating academic pressures, effective exam preparation, and managing logistics such as securing second-year accommodation.

## References

References available upon request.